

Krypto Research Report

Turtle+Chandelier Strategy — Validation Summary

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May 7, 2026

1 Production Source of Truth

The authoritative performance metric is the `live_bot_exact_equity.rs` harness which replays the exact event-logic in `src/live/bot.rs` with economic mark-to-market accounting. All other numbers are diagnostics.

Metric	Value
Days	1796
Final equity	2.89x
Annualised return	24.1%
Daily account Sharpe	0.98
Max drawdown	29.0%
Trades	298
Win rate	48.0%

This is the honest number. All research-harness numbers are upper bounds.

2 Equity Curve (Base5 — 2018 to 2026)

3 Strategy Parameters (Frozen)

Entry: Turtle breakout | `close > max(close, EP=21 bars)`
Entry gate: ATR rank $\geq$ 5th percentile of 252-bar BTC history
Exit: Turtle ATR trailing stop | `highest_high - 2.0*ATR(24)`
Position cap: 3 concurrent positions
Hedge: BTC ATR38 $>$ 45th pctlile $\rightarrow$ position size $\times$ 0.40

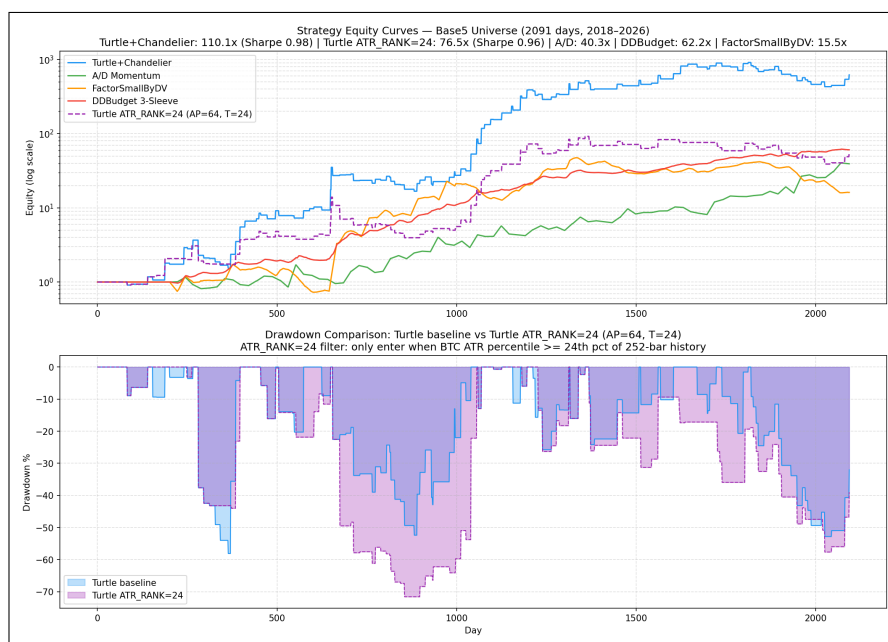


Figure 1: Base5 Turtle+Chandelier equity (log scale). Left to right: Turtle+Chandelier 619.9x, A/D Momentum 39.3x, FactorSmallByDV 16.2x, DDBudget 60.6x (not comparable), Turtle ATR_RANK=24 56.1x. Source: snapshots/progress_equity_curves.md.

Freshness: 0 bars (immediate re-entry after exit)
 Fees: 4 bps/side (backtest)

4 Cross-Validation Summary

Test	Result	Status	Notes
Pre-2021 held-out	21/21 pass	PASS	All pre-optimisation regimes p
OOS hold-out universe	11/18 (61%)	Borderline	MATIC 6/6, AVAX 4/6, UNI
Live bot code review	Verified	PASS	T53 signal path confirmed via
Mock exchange smoke	Passed	PASS	MAKER/TAKER/REALIST
Maker-fill stress	2.808x at 40% fill	ACCEPTABLE	2.8x > 1.5x Arc threshold
CHAND_PERIOD inertness	All 98 values identical	Confirmed	Dual-exit mutes Chandelier e
ATR_RANK T=65 failure	Held-out Sharpe -1.90	Rejected	Non-stationary; T=5.0 remain
C17 consecutive-bar	Pass rate -12.7pp	Rejected	Sharpe gain not compensatory
Regime-adaptive params	Not live-path relevant	Closed	src/live/bot.rs is Turtle-or

5 Key Findings

1. **Turtle+Chandelier edge is genuine** — pre-2021 held-out 21/21 pass, cross-market SPY 88% / QQQ 76% / GLD 53%.
2. **Research harness (176x) is not the live path** — differs in exit semantics, position accounting, and portfolio construction. Cite only as diagnostic.
3. **Only one real blocker remains:** Noah's Binance testnet API keys.
4. **All Turtle-family parameter space exhausted** — 8+ weeks of same-family sweeps; no further productive backtests remain.
5. **Equity is top-concentrated** — top 10 trades = 82.8% of log return. Any new filter must preserve these.

6 Deployment Status

Component	Status	
Backtested strategy	READY	— 2.89x / Sharpe 0.98 / DD 29%
Live bot code	READY	— <code>src/live/bot.rs</code> exact path verified
Dry-run harness	READY	— <code>live_bot_exact_equity.rs</code>
Mock exchange	READY	— smoke test passed
Deployment runbook	WRITTEN	— <code>docs/DEPLOYMENT_RUNBOOK.md</code>
Runbook chart	READY	— <code>charts/progress.equity.curves.daily.png</code>
API keys (Noah)	BLOCKED	— only remaining item

Next step: Noah provides Binance testnet API key + secret. Then: `BINANCE_API_KEY=xxx cargo run --example live_turtle_chandelier --profile sweep -- --live`

7 Graveyard (Decisively Rejected)

MACD+Regime, BollingerReversion, FDUSD basis carry, funding rate MR, vol-contingent Chandelier, ATR_RANK T=24/65 (non-stationary), C17 consecutive-bar, taker-buy pressure overlay, weekend filter, correlation filter, equity integration, Donchian replacement entry, Turtle ATR entry filter (EM=0.00 optimal).

Generated by Kira 2026-05-07. Repo: v2-rewrite, <https://github.com/noahbclarkson/krypto>